

Cryptography on the Internet

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ABSTRACT

Typical e-mail travels through many computers. Every one who uses computers can read, copy, and store this e-mail. Many of the competitors in the businesses are highly motivated to intercept e-mail. Sending information about one's business, legal, and personal mail through computers is even less confidential than sending the same material on a postcard.

Without security, the Internet is little more than the world's biggest bulletin board. The law enforcement agencies claim that security on the Internet should be outlawed because criminals use it to avoid detection.

Securing information on the Internet gives the average law abiding citizen a few of the privacy rights which governments, businesses, and other organizations need for their operations.

In this paper, I will discuss the security methods and requirement for the computer information on the Internet.

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INTRODUCTION

Should we trust the U.S. mail or the Federal Express? Is a telephone call a private conversation? Is an e-mail message safe? Some of these are safer than the others, but all of them carry some risk [1, 9, 13]. We need to convince a few million consumers that their credit card numbers can be safely sent over the Internet for commercial transactions. In fact, the security of digital information has been a major concern in the commercialization of the Internet[1, 2, 12].

We might know something about the important role of cryptography in World War II [1, 9]. While everyone has heard of the operations associated with the Central Intelligence Agency (CIA), many of us have never heard of the National Security Agency (NSA) [1, 9]. The NSA is responsible for developing secure communication technologies. Development of the best cryptography has always been a key component in the operation of the NSA [9].

Also, cryptography is an important issue for the client/server software developers, digital commerce, and all Internet users who want to keep their personal communications private [3, 8, 10, 13]. As the Internet becomes more commercialized, businesses need to provide secure transactions to alleviate the fears of consumers who are nervous about the risks of online shopping [1, 5, 9, 11, 12].

Today, there is a lot of sensitive personal information on the Internet which needs special protection. Online medical records must be handled with care so that employers and insurance companies can't review this. Cryptography lets businesses and people feel secure on the public network.

SINGLE-KEY

Single-key cryptography consists of the substitution of a different character for the original character. For example, the message "hello maria" will change to "funny lover" [2, 7]. Note that the letter "h" changes to "f", "e" change to "u", and so on. The process of creating a coded message is called encoding (encrypting), and the process of producing the original message is called decoding (decrypting) [2, 3, 7, 11, 13]. In a single-key cryptography, we use the same key for both encoding and decoding. It is easy to discover the key for a code by studying some encoded messages. One way is to identify the most frequently used characters and character sequences in the messages. For example, the letter "e" is the most frequently used letter in the English alphabet.

Also, the most frequently used three-letter word is "the". So, if the same three-letter character sequence is used over and over again, it might be "the". So the single-key cryptography is not very safe since many people must share the key.

DOUBLE-KEY

Double-key cryptography uses two keys, the first key for encoding messages and the second key for decoding messages [3, 7]. The first key is called "public key" and can be freely distributed to anyone. The other

sensitive information without authorization [1, 9, 12]..... Credit card records must be safe from newspaper reporters and the private investigators [1, 2, 9, 11, 12].

key is called "private key", and it is held by only one person.

A key is just a string of zeros and ones which is a string of bits. A good encryption method can be broken when one discovers the key. Of course, the longer string is more difficult to find out than the shorter string because there are more possibilities. There are 1,024 possible strings contained in 10 bits key, 1,048,576 possible strings containing with 20 bits key, and 1,099,511,627,776 possible strings containing 40 bits key. The 40-bit keys could be broken with a supercomputer in a several hours in 1995. Double-key cryptography is more sophisticated than single-key cryptography, but it is easy to discover the key [7, 9].

PRETTY GOOD PRIVACY

Pretty Good Privacy (PGP) is a high-security cryptography software application that allows people to exchange messages with privacy and authentication [14]. It encrypts and decrypts messages and provides protection for files and e-mail on the network. Although the initial program is written by Philip Zimmermann, other programmers around the globe have created subsequent PGP versions [9, 14]. PGP provides privacy for the politicians running election campaigns, taxpayers records, therapists clients' files, trade secrets, journalists' articles, and people seeking romance [9]. When using PGP, the program generates two keys. The first PGP key is called "private" which stays in the user's (creator of the key pair) computer [9]. The

second key is called “public” which is given to the correspondents. Let see an example of the sample public key:

```
BEGIN PGP PUBLIC KEY BLOCK
Version: 2.7
```

```
tuaFdfFDF343FdshfjdjfGFH3bdxcbx
jghsdritjgkgieutxdfdjfTTHgder
PhGsfdffbffhfnbnbcnvbTYUTbnbeqjwcbLQ
```

```
END PGP PUBLIC KEY BLOCK
```

Both the sender and receiver of the message should have the PGP program in their computer. The sender will send an e-mail and the above public key to the receiver. The receiver can store the above public key in his/her PGP program and use this public key to encrypt the e-mail message. When a person uses his/her correspondent’s public key, only the intended recipient can decrypt and read the message.

PGP could add the above public key to the “virtual key ring” for the user’s convenience [9]. A virtual key ring is a file that contains encryption keys. To add a key to the key ring file, one needs to know the name and e-mail address of the person associated with the key. Some of the commonly used PGP commands are as follows:

PGP -kg	Generate unique public/private key
PGP -ka	Add a public key to the key ring
PGP -e filename userid	Creates the binary file “filename.pgp”
PGP -ea filename userid	Creates the ASCII file “filename.asc”

PGP SIGNATURES

A PGP signature is a document validation device that operates as a safeguard against forgeries and alterations to an original document [4, 14, 15]. When a file or e-mail

message is signed with the PGP “digital signature”, this allows people who have PGP and the sender’s public key to verify that she/he wrote this document. Also, nobody has changed the content of this document. For example, you may see a PGP signature in an e-mail message, a Usenet article, or on a Web Page.

A PGP signature attached to a plain text file is called a **clear signature**, and a file signed with a clear signature is called **clear-signed** [9, 14, 15]. The following is an example of a clear-signed text file:

```
-- BEGIN PGP SIGNED MESSAGE--
10/10/98 voting results
```

```
Mehdi 79
Jeff 57
Maria 53
Roya 67
Mitra 71
Azita 70
```

```
-- BEGIN PGP SIGNATURE --
Version: 2.6.2
```

The PGP signature appears at the bottom by phrase such as “BEGIN PGP SIGNATURE.” Clear-signed documents are like an inverse of encrypted documents [9, 14]. We use a private key to create a signature and a public key to verify it.

LAW ENFORCEMENT AND ENCRYPTION

When very good cryptography is available for general use, then anyone can generate his/her own pair of keys, post a public key in a public directory, and exchange encrypted e-mail with anyone else [1, 6]. This means that terrorists can safely communicate with one another and say anything without fear of discovery. Criminals can plan robberies and plot murders.

Powerful cryptography on the Internet pulls the law enforcement plug in a big way [1, 9]. As long as the codes can't be cracked, there is no way for the FBI to monitor an encrypted e-mail conversation between suspected criminals [1, 9]. Drug dealers, organized crime, terrorist organizations, and random criminals can all use strong encryption to great advantage. People who use PGP today routinely create 1,024-bit private keys, thereby ensuring a level of security like that of top -secret military communications.

What can the government hope to do about this technology? One possibility is to outlaw the use of strong cryptography. Cryptography software would be restricted and distributed only to authorized personnel, probably through some sort of government licensing. There are problems with this, however. Some people would oppose laws outlawing strong cryptography based, at least in part, on the First Amendment [1, 9]. Another problem is the global nature of the Internet [1, 4, 5, 8, 9]. A law that applies to U.S. citizens would not be enforceable for messages exchanged by non-U.S. citizens. We may be able to prohibit our own citizens from using strong cryptography, but how can we hope to control the rest of the world's population?

Another possible solution is that government controls the digital communication with the cooperation of major hardware manufacturers in the absence of legislation and public debate [9]. Double-key cryptography would be hard-wired into all computer chips inserted into modems, telephones, and other network communication devices. Then, any computer communication over the Internet would have to pass through a government-approved encryption chip. Each encryption

chip would be programmed with its own unique key pair.

To give law enforcement the power to monitor encrypted communications, the government would need access to all of the private keys in all of the encryption chips. Whenever an encryption chip was manufactured, a copy of its private key would be handed to the government for safekeeping and indexed by its serial number [9]. No one would have access to any of these private keys except under a court order authorizing a digital wiretap. Then, the serial numbers of the relevant devices would be used to retrieve the necessary private keys for the proper authorities. This would effectively give law enforcement personnel a "back door" into all Internet communications.

Phil was profoundly bothered by the possibility that privacy might become the property of a federal agency [4, 9]. The resulting PGP package made it possible for users to generate key pairs for strong encryption. It also gave them a lot of convenient utilities for encoding, decoding, and managing collections of PGP keys on virtual key rings [4, 9].

Massachusetts Institute of Technology (MIT) is the official distribution site for PGP [4, 9]. Downloading PGP from MIT involves the most complicated downloading procedure on the Internet. If you are not a U.S. or Canadian citizen, you cannot legally download it [9]. MIT does not police the distribution of PGP, but they do what they need to do to distribute PGP without violating the law [9]. This presumably includes maintaining a record of all Internet addresses receiving PGP downloads.

Although the original PGP software was written for UNIX, MIT now distributes

versions for Mac's and DOS platforms as well [9, 14]. Other sources distribute PGP for PC Windows [9].

CONCLUSION

Phil Zimmermann often explains PGP via an analogy involving letters and postcards. He argues that encrypted e-mail is like a letter in an envelope. An e-mail message that is not encrypted is like a message on a postcard. Anyone can read what you write on the back of a postcard. The Postmaster General can open envelopes containing letters when authorized by a court order. PGP is not like a letter in an envelope because no one can inspect a PGP file unless they hold the one and only private key that can decode the file.

As soon as the commercial venders feel that it's safe to build PGP into their products, strong encryption is likely to be integrated into all network communication packages. Phil Zimmermann appears to be betting on the eventual commercialization of PGP. In 1996, shortly after the Grand Jury halted its investigation, he founded his own company to develop enhanced versions of PGP with easy-to-use GUIs.

Only time will tell if the digital world will some day be permeated by trap doors. In the meantime, PGP is still subject to export restrictions, and one is well-advised to honor the legal restrictions that pertain to PGP.

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